PRAKTIKUM BASIS DATA

MODUL 08

Scripting SQL (Create index, Create view, Analyze Table)

Oleh:

(Quratu Ayun Defaren) (103122400064)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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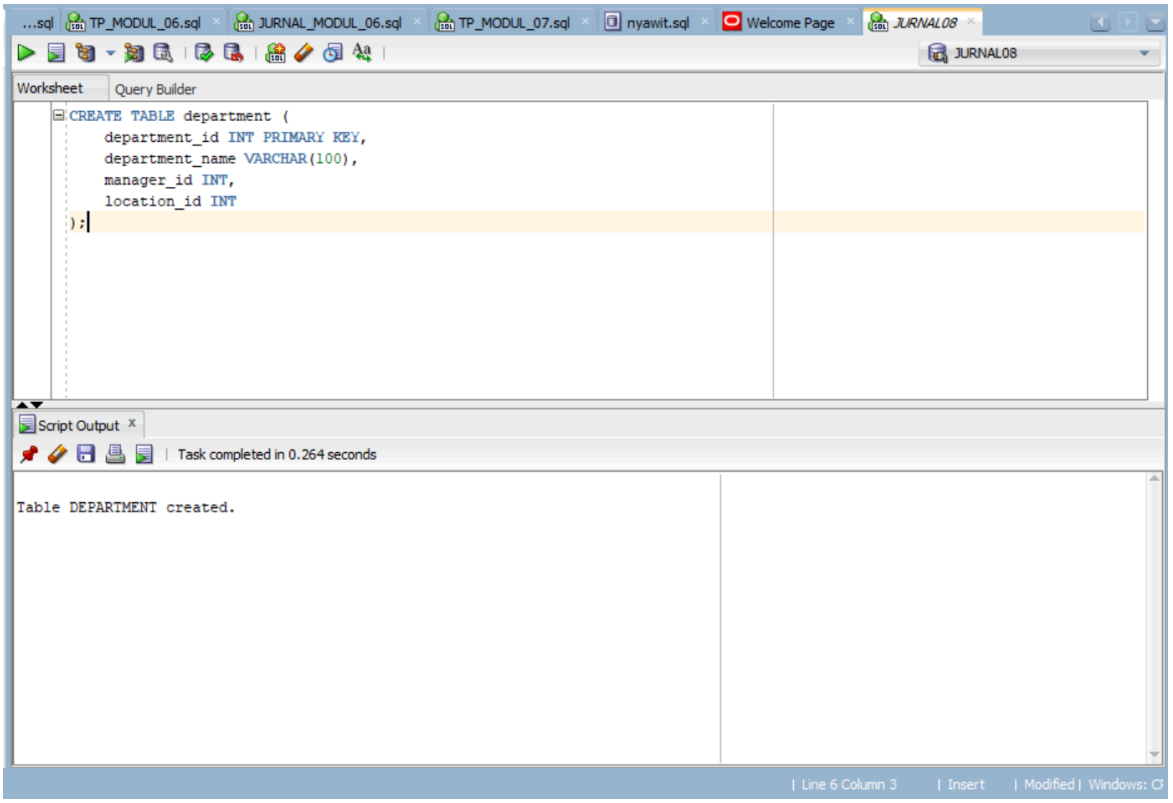
JURNAL

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

1. Desain Database

a.



b.

```sql
CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10,2),
    commission_pct DECIMAL(5,2),
    hire_date DATE NOT NULL,
    department_id INT,

    CONSTRAINT fk_dept
    FOREIGN KEY (department_id)
    REFERENCES department(department_id)
);
```

```
Table DEPARTMENT created.

Table EMPLOYEE created.
```

c.

```sql
    FOREIGN KEY (supplier_id)
    REFERENCES supplier(supplier_id)
);

CREATE TABLE inventory_transaction (
    transaction_id INT PRIMARY KEY,
    product_id INT,
    quantity INT,
    transaction_date DATE,

    FOREIGN KEY (product_id)
    REFERENCES product(product_id)
);
```

```
Table EMPLOYEE created.

Table SUPPLIER created.

Table PRODUCT created.

Table INVENTORY_TRANSACTION created.
```

2.

```sql
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE product COMPUTE STATISTICS;
ANALYZE TABLE inventory_transaction COMPUTE STATISTICS;
```

Script Output — Task completed in 0.096 seconds

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.
```

3.

```sql
INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'IT', 102, 1002);
INSERT INTO department VALUES (30, 'Finance', 103, 1003);
INSERT INTO department VALUES (40, 'Marketing', 104, 1004);
INSERT INTO department VALUES (50, 'Sales', 105, 1005);
INSERT INTO department VALUES (60, 'Logistic', 106, 1006);
INSERT INTO department VALUES (70, 'Support', 107, 1007);
INSERT INTO department VALUES (80, 'Engineering', 108, 1008);
INSERT INTO department VALUES (90, 'QA', 109, 1009);
INSERT INTO department VALUES (100, 'Admin', 110, 1010);
```

Script Output — Task completed in 0.13 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Power & Battery

Your battery is low
You should plug in your device.

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

JURNAL08

Worksheet | Query Builder

```sql
INSERT INTO employee VALUES (1, 'Andi', 'andi@mail.com', 5000, 0.1, TO_DATE('2022-01-01', 'YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (2, 'Budi', 'budi@mail.com', 6000, 0.2, TO_DATE('2022-02-01','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (3, 'Citra', 'citra@mail.com', 5500, 0.15, TO_DATE('2022-03-01','YYYY-MM-DD'), 10);
INSERT INTO employee VALUES (4, 'Dedi', 'dedi@mail.com', 7000, 0.25, TO_DATE('2022-04-01','YYYY-MM-DD'), 20);
INSERT INTO employee VALUES (5, 'Eka', 'eka@mail.com', 6500, 0.1, TO_DATE('2022-05-01','YYYY-MM-DD'), 30);
INSERT INTO employee VALUES (6, 'Fajar', 'fajar@mail.com', 6200, 0.12, TO_DATE('2022-06-01','YYYY-MM-DD'), 40);
INSERT INTO employee VALUES (7, 'Gina', 'gina@mail.com', 5800, 0.13, TO_DATE('2022-07-01','YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 7200, 0.2, TO_DATE('2022-08-01','YYYY-MM-DD'), 60);
INSERT INTO employee VALUES (9, 'Intan', 'intan@mail.com', 7100, 0.18, TO_DATE('2022-09-01','YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (10, 'Joko', 'joko@mail.com', 8000, 0.3, TO_DATE('2022-10-01','YYYY-MM-DD'), 80);
```

Script Output | Task completed in 0.174 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Line 72 Column 35 | Insert | Modified | Windows: CF

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

JURNAL08

Worksheet | Query Builder

```sql
INSERT INTO supplier VALUES (1, 'Supplier A', '0811111111');
INSERT INTO supplier VALUES (2, 'Supplier B', '0822222222');
INSERT INTO supplier VALUES (3, 'Supplier C', '0833333333');
INSERT INTO supplier VALUES (4, 'Supplier D', '0844444444');
INSERT INTO supplier VALUES (5, 'Supplier E', '0855555555');
INSERT INTO supplier VALUES (6, 'Supplier F', '0866666666');
INSERT INTO supplier VALUES (7, 'Supplier G', '0877777777');
INSERT INTO supplier VALUES (8, 'Supplier H', '0888888888');
INSERT INTO supplier VALUES (9, 'Supplier I', '0899999999');
INSERT INTO supplier VALUES (10, 'Supplier J', '0800000000');
```

Script Output | Task completed in 0.107 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Line 88 Column 1 | Insert | Modified | Windows: CF

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

JURNAL08

Worksheet | Query Builder

```sql
INSERT INTO product VALUES (1, 'Laptop', 10000000, 10, 1);
INSERT INTO product VALUES (2, 'Mouse', 200000, 50, 2);
INSERT INTO product VALUES (3, 'Keyboard', 300000, 40, 3);
INSERT INTO product VALUES (4, 'Monitor', 2000000, 20, 4);
INSERT INTO product VALUES (5, 'Printer', 1500000, 15, 5);
INSERT INTO product VALUES (6, 'Scanner', 1200000, 10, 6);
INSERT INTO product VALUES (7, 'Speaker', 500000, 25, 7);
INSERT INTO product VALUES (8, 'Webcam', 400000, 30, 8);
INSERT INTO product VALUES (9, 'Router', 800000, 18, 9);
INSERT INTO product VALUES (10, 'Flashdisk', 100000, 100, 10);
```

Script Output | Task completed in 0.156 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

...sql | TP_MODUL_06.sql | JURNAL_MODUL_06.sql | TP_MODUL_07.sql | nyawit.sql | Welcome Page | JURNAL08

JURNAL08

Worksheet | Query Builder

```sql
INSERT INTO product VALUES (10, 'Flashdisk', 100000, 100, 10);

INSERT INTO inventory_transaction VALUES (1, 1, 5, TO_DATE ('2024-01-01', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (2, 2, 10, TO_DATE ('2024-01-02', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (3, 3, 8, TO_DATE ('2024-01-03', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (4, 4, 6, TO_DATE ('2024-01-04', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (5, 5, 7, TO_DATE ('2024-01-05', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (6, 6, 4, TO_DATE ('2024-01-06', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (7, 7, 9, TO_DATE ('2024-01-07', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (8, 8, 3, TO_DATE ('2024-01-08', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (9, 9, 2, TO_DATE ('2024-01-09', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));
```

Script Output | Task completed in 0.118 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

| Line 105 Column 25 | Insert | Modified | Windows: CF

4.

```
    REFERENCES product(product_id)
);

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE product COMPUTE STATISTICS;
ANALYZE TABLE inventory_transaction COMPUTE STATISTICS;

INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'IT', 102, 1002);
INSERT INTO department VALUES (30, 'Finance', 103, 1003);
```

Script Output — Task completed in 0.108 seconds

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.
```

5.

```
INSERT INTO inventory_transaction VALUES (8, 8, 3, TO_DATE ('2024-01-08', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (9, 9, 2, TO_DATE ('2024-01-09', 'YYYY-MM-DD'));
INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

Script Output — Task completed in 0.082 seconds

```
Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

Table PRODUCT analyzed.

Table INVENTORY_TRANSACTION analyzed.

View EMPVU80 created.
```

6.

```sql
INSERT INTO inventory_transaction VALUES (10, 10, 15, TO_DATE ('2024-01-10', 'YYYY-MM-DD'));

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;
```

```
Table PRODUCT analyzed.


Table INVENTORY_TRANSACTION analyzed.


View EMPVU80 created.

Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER(38)
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER(38)
```

7.

```sql
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;

CREATE INDEX idx_employee_dept
ON employee(department_id);
```

```
Table INVENTORY_TRANSACTION analyzed.


View EMPVU80 created.

Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER(38)
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER(38)

Index IDX_EMPLOYEE_DEPT created.
```